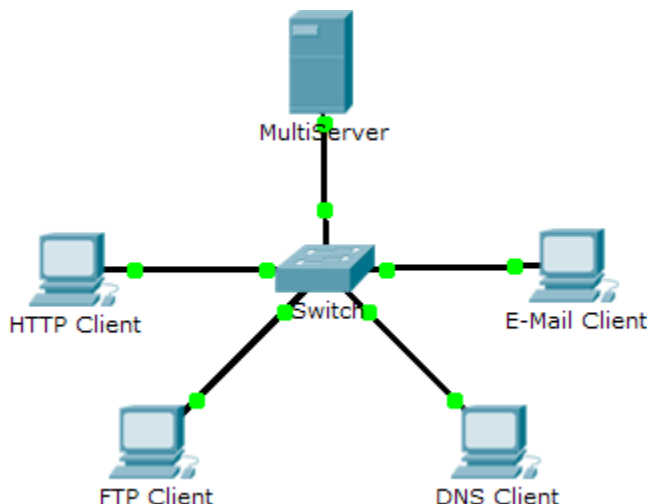


Packet Tracer Simulation - TCP and UDP Communications

Topology



Objectives

Part 1: Generate Network Traffic in Simulation Mode

Part 2: Examine the Functionality of the TCP and UDP Protocols

Background

This simulation activity is intended to provide a foundation for understanding the TCP and UDP in detail. Simulation mode provides the ability to view the functionality of the different protocols.

As data moves through the network, it is broken down into smaller pieces and identified in some fashion so that the pieces can be put back together. Each of these pieces is assigned a specific name (protocol data unit [PDU]) and associated with a specific layer. Packet Tracer Simulation mode enables the user to view each of the protocols and the associated PDU. The steps outlined below lead the user through the process of requesting services using various applications available on a client PC.

This activity provides an opportunity to explore the functionality of the TCP and UDP protocols, multiplexing and the function of port numbers in determining which local application requested the data or is sending the data.

Part 1: Generate Network Traffic in Simulation Mode

Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

- Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.
- Enter the **ping 192.168.1.255** command. This will take a few seconds as every device on the network responds to **MultiServer**.
- Close the **MultiServer** window.

Step 2: Generate web (HTTP) traffic.

- a. Switch to Simulation mode.
- b. Click **HTTP Client** and click the **Desktop** tab > **Web Browser**.
- c. In the URL field, enter **192.168.1.254** and click **Go**. Envelopes (PDUs) will appear in the simulation window.
- d. Minimize, but do not close, the **HTTP Client** configuration window.

Step 3: Generate FTP traffic.

- a. Click **FTP Client** and click the **Desktop** tab > **Command Prompt**.
- b. Enter the **ftp 192.168.1.254** command. PDUs will appear in the simulation window.
- c. Minimize, but do not close, the **FTP Client** configuration window.

Step 4: Generate DNS traffic.

- a. Click **DNS Client** and click the **Desktop** tab > **Command Prompt**.
- b. Enter the **nslookup multiserver.pt.ptu** command. A PDU will appear in the simulation window.
- c. Minimize, but do not close, the **DNS Client** configuration window.

Step 5: Generate Email traffic.

- a. Click **E-Mail Client** and click the **Desktop** tab > **E Mail** tool.
- b. Click **Compose** and enter the following information:
 - 1) **To:** user@multiserver.pt.ptu
 - 2) **Subject:** Personalize the subject line
 - 3) **E-Mail Body:** Personalize the Email
- c. Click **Send**.
- d. Minimize, but do not close, the **E-Mail Client** configuration window.

Step 6: Verify that the traffic is generated and ready for simulation.

Every client computer should have PDUs listed in the Simulation Panel.

Part 2: Examine Functionality of the TCP and UDP Protocols

Step 1: Examine multiplexing as all of the traffic crosses the network.

You will now use the **Capture/Forward** button and the **Back** button in the Simulation Panel.

- a. Click **Capture/Forward** once. All of the PDUs are transferred to the switch.
- b. Click **Capture/Forward** again. Some of the PDUs disappear. What do you think happened to them?
Some PDUs are consumed by the switch, which processes them and forwards them to the appropriate destination. This action reflects the process of multiplexing.
- c. Click **Capture/Forward** six times. All clients should have received a reply. Note that only one PDU can cross a wire in each direction at any given time. What is this called?
This is called duplex communication which only allow one PDU cross the wire in either direction at any given time.

- d. A variety of PDUs appears in the event list in the upper right pane of the simulation window. Why are they so many different colors?

Each protocol is color-coded for easy identification. So different colors represent different type of traffic.

Green- Indicates successful communication or traffic flow without issues.

Purple- Represents application-level data or specific services.

Brown- Indicates network-level traffic or intermediate routing.

Grey- Often used to represent packets that are being dropped or are incomplete.

- e. Click **Back** eight times. This should reset the simulation.

Note: Do not click **Reset Simulation** any time during this activity; if you do, you will need to repeat the steps in Part 1.

Step 2: Examine HTTP traffic as the clients communicate with the server.

- a. Filter the traffic that is currently displayed to display only **HTTP** and **TCP** PDUs filter the traffic that is currently displayed:
- 1) Click **Edit Filters** and toggle the **Show All/None** check box.
 - 2) Select **HTTP** and **TCP**. Click anywhere outside of the Edit Filters box to hide it. The Visible Events should now display only **HTTP** and **TCP** PDUs.
- b. Click **Capture/Forward**. Hold your mouse above each PDU until you find one that originates from **HTTP Client**. Click the PDU envelope to open it.
- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

TCP

Are these communications considered to be reliable?

Yes, TCP is considered reliable because it ensures data is delivered accurately and in order through mechanisms like connection establishment (three-way handshake), acknowledgment of received data, sequencing to reorder packets, error detection and correction, and retransmission of lost packets

- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

SRC PORT - 1025

DEST PORT – 80

SEQUENCE NUM – 0

ACK NUM – 0

The field to the left of the **WINDOW** field is **FLAGS** with the value of 0b00000010

TCP			
0	4	8	16
SOURCE PORT:1025		DESTINATION PORT 80	
SEQUENCE NUMBER 0			
ACKNOWLEDGEMENT NUMBER 0			
OFFSET 0x0	RESERVED 0	FLAGS 0b0000001 0	WINDOW 65535
CHECKSUM 0x0000		URGENT POINTER 0x0000	
OPTION			PADDING: 0
DATA (VARIABLE LENGTH)			

- e. Close the PDU and click **Capture/Forward** until a PDU returns to the **HTTP Client** with a checkmark.
- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers

different than before?

SRC PORT – 80

DEST PORT – 1025

SEQ NUM – 0

The source and destination ports swapped because the server is now replying to the client using the same TCP session. The sequence number remains 0 because no new data is being sent with this reply packet—only an acknowledgment. The acknowledgment number increases to 1 to confirm receipt of the client's first packet and signal readiness for the next byte.

TCP									
0 4 8 16 24 Bytes									
SOURCE PORT 80					DESTINATION PORT 1025				
SEQUENCE NUMBER 0									
ACKNOWLEDGEMENT NUMBER 1									
OFFSET 0x0	RESER VED: 0	FLAGS 0x001101			WINDOW 16384				
CHECKSUM 0x0000					URGENT POINTER 0x0000				
OPTION									
DATA (VARIABLE LENGTH)					PADDING: 0				

- g. There is a second **PDU** of a different color, which **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the HTTP communication. Click this second PDU envelope and select **Outbound PDU Details**.
- h. What information is now listed in the TCP section? How are the port and sequence numbers different from the previous two PDUs?

TCP										0	4	8	16	24	Bytes
SOURCE PORT 1025					DESTINATION PORT 80										
SEQUENCE NUMBER 1															
ACKNOWLEDGEMENT NUMBER 1															
OFFSET 0x0	RESERVED 0	FLAGS 0x001100			WINDOW 65535										
CHECKSUM 0x0000					URGENT POINTER 0x0000										
OPTION															
DATA (VARIABLE LENGTH)							PADDING 0								
HTTP REQUEST															
0 4 8 16 Bytes															
HTTP Data: Accept-Language: en-us Accept: */*															

The source port and destination port are still the same because in the same session. The sequence number will now increase and represent the starting byte of the HTTP request data being sent. This packet carries actual data, so the sequence number advances from 0. The acknowledgment number reflects the server's sequence number, ensuring the communication remains synchronized.

- i. Click **Back** until the simulation is reset.

Step 3: Examine FTP traffic as the clients communicate with the server.

- a. In the Simulation Panel, change **Edit Filters** to display only **FTP** and **TCP**.
- b. Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **FTP Client**. Click that PDU envelope to open it.
- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

TCP

Are these communications considered to be reliable?

Yes, TCP is considered reliable because it ensures data is delivered accurately and in order through mechanisms like connection establishment (three-way handshake), acknowledgment of received data, sequencing to reorder packets, error detection and correction, and retransmission of lost packets

- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

TCP		Bin	
SOURCE PORT 1025		DESTINATION PORT 21	
SEQUENCE NUMBER 0			
ACKNOWLEDGEMENT NUMBER 0			
OFFSET 0	RESERVED 0	FLAGS 0x00000001	WINDOW 65535
CHECKSUM 0x0000		URGENT POINTER 0x0000	
OPTION			PADDING: 0
DATA (VARIABLE LENGTH)			

SRC PORT - 1025

DEST PORT – 21

SEQUENCE NUM – 0

ACK NUM – 0

The field to the left of the WINDOW field is FLAGS with the value of 0b000000010

- e. Close the PDU and click **Capture/Forward** until a PDU returns to the **FTP Client** with a checkmark.
 f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

TCP				Bin			
3		4		16		24	
SOURCE PORT 21				DESTINATION PORT 1025			
SEQUENCE NUMBER 0							
ACKNOWLEDGEMENT NUMBER 1							
OFFSET 0		RESERVED 0		FLAGS 00001001		WINDOW 1024	
CHECKSUM 0x0000				URGENT POINTER 0x0000			
OPTION							
DATA (VARIABLE LENGTH)						PADDING 0	

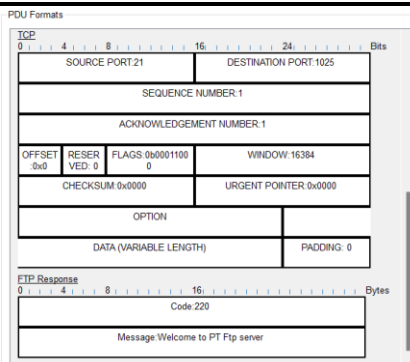
The source and destination ports swapped because the server is now replying to the client using the same TCP session. The sequence number remains 0 because no new data is being sent with this reply packet—only an acknowledgment. The acknowledgment number increases to 1 to confirm receipt of the client's first packet and signal readiness for the next byte.

- g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?

TCP		Bin			
4 8 16 24					
SOURCE PORT 1025				DESTINATION PORT 21	
SEQUENCE NUMBER 1					
ACKNOWLEDGEMENT NUMBER 1					
OFFSET 0x0	RESERVED 0	FLAGS 00001000	WINDOW 65535		
CHECKSUM 0x0000		URGENT POINTER 0x0000			
OPTION			PADDING 0		
DATA (VARIABLE LENGTH)					

The source port and destination port are still the same because in the same session. The sequence number will now increase and represent the starting byte of the HTTP request data being sent. This packet carries actual data, so the sequence number advances from 0. The acknowledgment number reflects the server's sequence number, ensuring the communication remains synchronized.

- h. Close the PDU and click **Capture/Forward** until a second PDU returns to the **FTP Client**. The PDU is a different color.
 i. Open the PDU and select **Inbound PDU Details**. Scroll down past the TCP section. What is the message from the server?



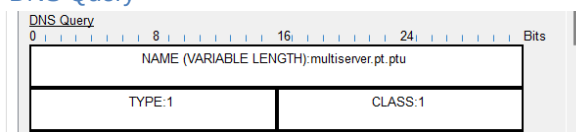
The message from the server is a FTP Response with 2 fields which includes Code with value of 220 and Message with "Welcome to PT Ftp server".

- j. Click **Back** until the simulation is reset.

Step 4: Examine DNS traffic as the clients communicate with the server.

- a. In the Simulation Panel, change **Edit Filters** to display only **DNS** and **UDP**.
- b. Click the PDU envelope to open it.
- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

DNS Query



Are these communications considered to be reliable?

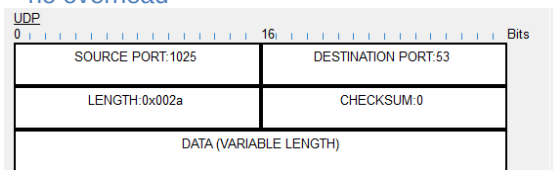
UDP (User Datagram Protocol) is considered unreliable because it does not establish a connection before sending data and does not provide mechanisms to ensure transmission, sorting, or error correction. There is no sequence or acknowledgement number in UDP.

- d. Record the **SRC PORT** and **DEST PORT** values. Why is there no sequence and acknowledgement number?

SOURCE PORT – 1025

DEST PORT – 53

There is no sequence or acknowledgement number in UDP because it does not track the delivery of packets or maintain the state of the connection - its purpose is to pass data as quickly as possible with no overhead



- e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **DNS Client** with a checkmark.
- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

The Source Port and Destination Port value swapped, because the communication direction changes since the server is responding to client.

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UDP	
0	16
SOURCE PORT:53	DESTINATION PORT:1025
LENGTH:0x004a	CHECKSUM:0
DATA (VARIABLE LENGTH)	

- g. What is the last section of the **PDU** called?

DNS Answer

DNS Answer	
0	16
NAME (VARIABLE LENGTH):multiserver.pt.ptu	
TYPE:1	CLASS:1
TTL:86400	
LENGTH:4	IP:192.168.1.254

- h. Click **Back** until the simulation is reset.

Step 5: Examine email traffic as the clients communicate with the server.

- a. In the Simulation Panel, change **Edit Filters** to display only **POP3**, **SMTP** and **TCP**.

- b. Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **E-mail Client**. Click that PDU envelope to open it.
- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What transport layer protocol does email traffic use?

TCP

Are these communications considered to be reliable?

Yes, TCP is considered reliable because it ensures data is delivered accurately and in order through mechanisms like connection establishment (three-way handshake), acknowledgment of received data, sequencing to reorder packets, error detection and correction, and retransmission of lost packets.

- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

TCP																							
0				4				8				16				24				Bits			
SOURCE PORT 1025								DESTINATION PORT 25															
SEQUENCE NUMBER 0																							
ACKNOWLEDGEMENT NUMBER 0																							
OFFSET 0x0				RESERVED 0				FLAGS 0x00000001 0				WINDOW 65535											
CHECKSUM 0x0000								URGENT POINTER 0x0000															
OPTION																							
DATA (VARIABLE LENGTH)												PADDING: 0											

SRC PORT - 1025

DEST PORT - 25

SEQUENCE NUM - 0

ACK NUM - 0

The field to the left of the WINDOW field is FLAGS with the value of 0b00000001

- e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **E-Mail Client** with a checkmark.
- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

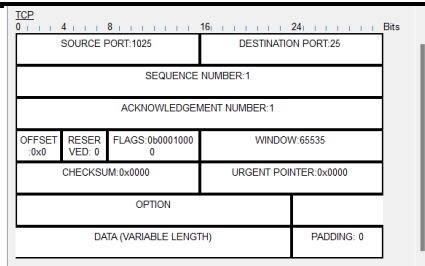
TCP																							
0				4				8				16				24				Bits			
SOURCE PORT: 25								DESTINATION PORT: 1025															
SEQUENCE NUMBER: 0																							
ACKNOWLEDGEMENT NUMBER: 1																							
OFFSET: 0x0				RESERVED: 0				FLAGS: 0x0001001				WINDOW: 10384											
CHECKSUM: 0x0000								URGENT POINTER: 0x0000															
OPTION																							
DATA (VARIABLE LENGTH)																PADDING: 0							

The source and destination ports swapped because the server is now replying to the client using the same TCP session. The sequence number remains 0 because no new data is being sent with this reply packet—only an acknowledgment. The acknowledgment number increases to 1 to confirm receipt of the client's first packet and signal readiness for the next byte.

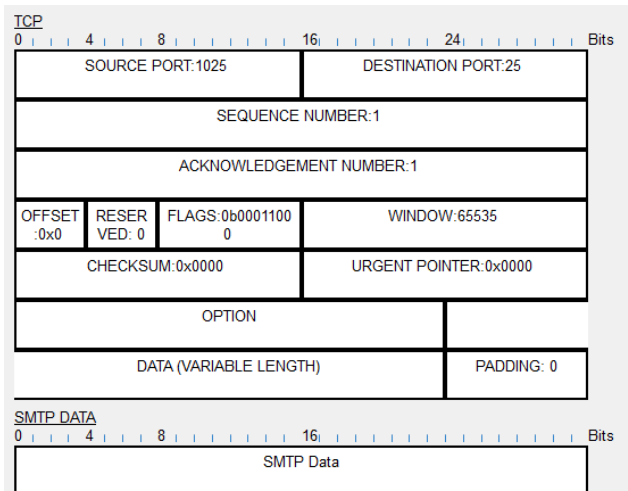
- g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?

The source port and destination port are still the same because in the same session. The sequence number will now increase and represent the starting byte of the HTTP request data being sent. This packet carries actual data, so the sequence number advances from 0. The acknowledgment number reflects the server's sequence number, ensuring the communication remains synchronized.

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- h. There is a second **PDU** of a different color that **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the email communication. Click this second PDU envelope and select **Outbound PDU Details**.



- i. How are the port and sequence numbers different from the previous two **PDU**s?

SOURCE PORT – 1025

DEST PORT – 25

SEQUENCE NUMBER – 1

ACK NUMBER – 1

SCR PORT and DEST PORT are reversed, SEQUENCE NUM and ACK NUM changed to 1, and FLAGS changed to PSH + ACK

- j. What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?
 TCP Port 25: The email protocol associated with TCP port 25 is SMTP (Simple Mail Transfer Protocol). SMTP is used for sending emails from clients to servers and between mail servers.

TCP Port 110: The email protocol associated with TCP port 110 is POP3 (Post Office Protocol version 3). POP3 is used for retrieving emails from a mail server to a client.

- k. Click **Back** until the simulation is reset.

Step 6: Examine the use of port numbers from the server.

- a. To see TCP active sessions, perform the following steps in quick succession:
- 1) Switch back to **Realtime** mode.
 - 2) Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.
- b. Enter the **netstat** command. What protocols are listed in the left column?

What port numbers are being used by the server?

TCP

Local Address: 192.168.1.254:21

Foreign Address: 192.168.1.2:1025

The :21 in the local address corresponds to the FTP service.

Port 1025 in the foreign address is a dynamically assigned client port used for this specific session.

```
C:\>netstat

Active Connections

    Proto Local Address          Foreign Address         State
    TCP    192.168.1.254:21      192.168.1.2:1025      ESTABLISHED
C:\>
```

- c. What states are the sessions in?

ESTABLISHED

- d. Repeat the **netstat** command several times until you see only one session still ESTABLISHED. For which service is this connection still open?

The connection still open is for the FTP (File Transfer Protocol) service.

Why doesn't this session close like the other three? (Hint: Check the minimized clients)

The client is still open or actively waiting for further communication. Unlike other services, FTP often requires multiple interactions between the client and the server during an ongoing session, so the session does not close immediately.

Suggested Scoring Rubric

Activity Section	Question Location	Possible Points	Earned Points
Part 2: Examine Functionality of the TCP and UDP Protocols	Step 1	15	
	Step 2	15	
	Step 3	15	
	Step 4	15	
	Step 5	15	
	Step 6	25	
Total Score		100	